OpenAPI Webhooks

There are cases where you want to tell your API **users** that your app could call *their* app (sending a request) with some data, normally to **notify** of some type of **event**.

This means that instead of the normal process of your users sending requests to your API, it's **your API** (or your app) that could **send requests to their system** (to their API, their app).

This is normally called a **webhook**.

Webhooks steps

The process normally is that **you define** in your code what is the message that you will send, the **body of the request**.

You also define in some way at which **moments** your app will send those requests or events.

And **your users** define in some way (for example in a web dashboard somewhere) the **URL** where your app should send those requests.

All the **logic** about how to register the URLs for webhooks and the code to actually send those requests is up to you. You write it however you want to in **your own code**.

Documenting webhooks with FastAPI and OpenAPI

With **FastAPI**, using OpenAPI, you can define the names of these webhooks, the types of HTTP operations that your app can send (e.g. `POST`, `PUT`, etc.) and the request **bodies** that your app would send.

This can make it a lot easier for your users to **implement their APIs** to receive your **webhook** requests, they might even be able to autogenerate some of their own API code.

> **Info**
>
> Webhooks are available in OpenAPI `3.1.0` and above, supported by FastAPI `0.99.0` and above.

An app with webhooks

When you create a **FastAPI** application, there is a `webhooks` attribute that you can use to define *webhooks*, the same way you would define *path operations*, for example with `@app.webhooks.post()`.

Python 3.10+

```python
from datetime import datetime

from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI()


class Subscription(BaseModel):
```

Ask AI

```
    username: str
    monthly_fee: float
    start_date: datetime


@app.webhooks.post("new-subscription")
def new_subscription(body: Subscription):
    """
    When a new user subscribes to your service we'll send you a POST request with this
    data to the URL that you register for the event `new-subscription` in the dashboard.
    """


@app.get("/users/")
def read_users():
    return ["Rick", "Morty"]
```

The webhooks that you define will end up in the **OpenAPI** schema and the automatic **docs UI**.

> **Info**
>
> The `app.webhooks` object is actually just an `APIRouter`, the same type you would use when structuring your app with multiple files.

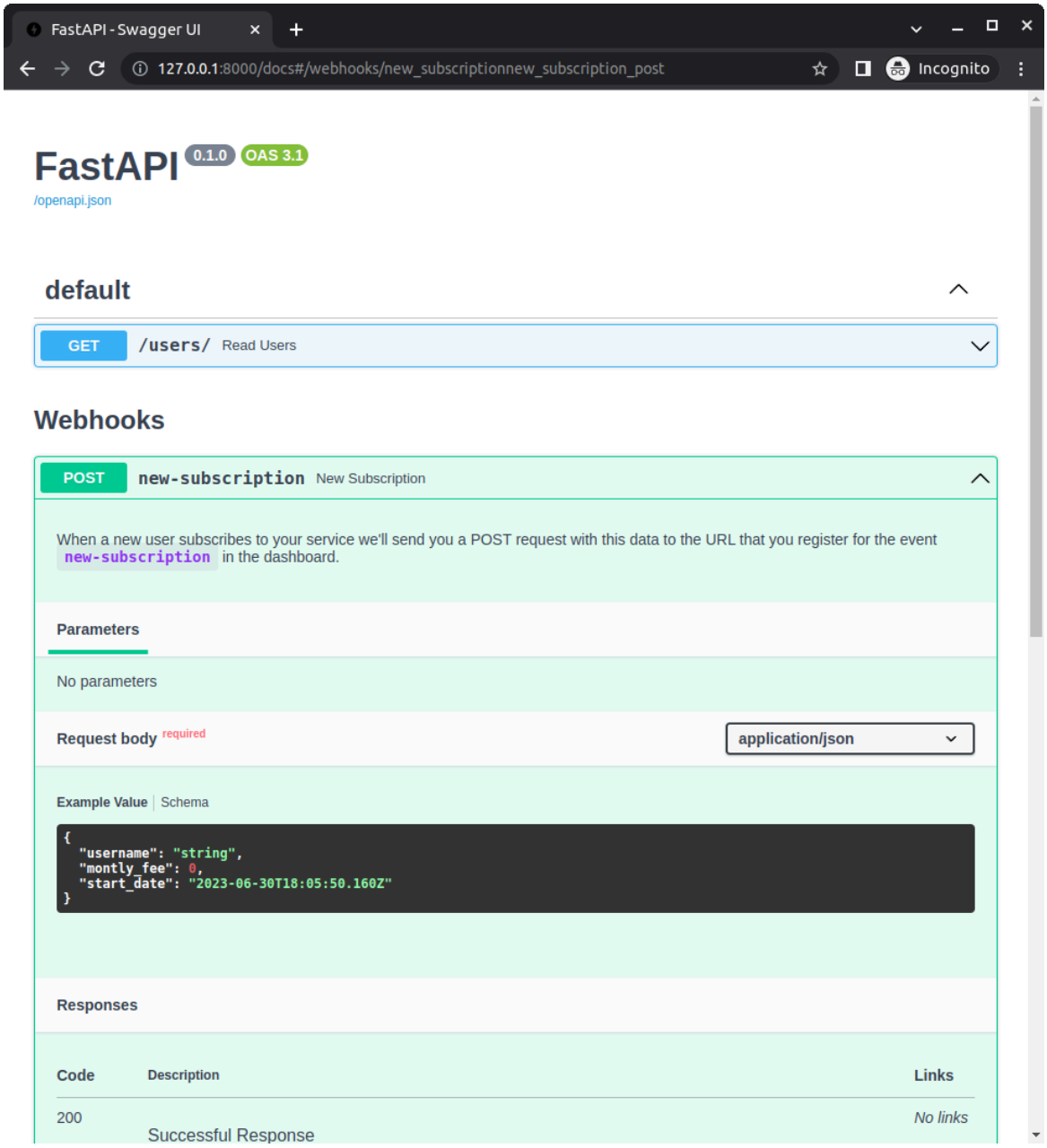
Notice that with webhooks you are actually not declaring a *path* (like `/items/`), the text you pass there is just an **identifier** of the webhook (the name of the event), for example in `@app.webhooks.post("new-subscription")`, the webhook name is `new-subscription`.

This is because it is expected that **your users** would define the actual **URL path** where they want to receive the webhook request in some other way (e.g. a web dashboard).

Check the docs

Now you can start your app and go to http:// 1 2 7 . 0 . 0 . 1 : 8 0 0 0 /docs.

You will see your docs have the normal *path operations* and now also some **webhooks**:


FastAPI - Swagger UI
127.0.0.1:8000/docs#/webhooks/new_subscriptionnew_subscription_post
Incognito
FastAPI
0.1.0
OAS 3.1
/openapi.json
default
GET
/users/
Read Users
Webhooks
POST
new-subscription
New Subscription
When a new user subscribes to your service we'll send you a POST request with this data to the URL that you register for the event new-subscription in the dashboard.
Parameters
No parameters
Request body
required
application/json
Example Value
Schema
{
"username": "string",
"montly_fee": 0,
"start_date": "2023-06-30T18:05:50.160Z"
}
Responses
Code
Description
Links
200
Successful Response
No links